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Conceptual understanding of Semiconductor in Analog Electronics.

1. In n-type (negative) material, a pentavalent impurity like phosphorus is added, which has five valence electrons, one more than Silicon, creating a surplus of free electrons that act as a majority charge carriers while in p-type (positive) material, a trivalent impurity like boron is ~~added~~ added, which has only three valence electrons, creating an empty space or hole in the crystal structure that behaves like a positive charge and acts as the majority carrier while the n-type donates extra electrons to the material, the p-type accepts them into these holes.

2, In a p-n junction diode, forward bias is established when ~~the~~ the p-type side is connected to the positive terminal of a power supply and the n-type side to the negative terminal. This reduces the barrier potential and narrows the depletion region allowing charge carriers to cross the junction easily, so a significant current flows through the diode. In reverse bias, the p-type side is connected to the negative terminal and n-type side to the positive terminal which increases the barrier potential and widens the depletion region, preventing charge carriers from crossing the junction, as a result, only a very small leakage current flows until breakdown occurs at high reverse voltage.

3, Ideal as applied to a device or a system means a perfect, simplified model that behaves exactly as expected without any losses, limitations, or unwanted effects. An ideal device operates with maximum efficiency and accuracy, ignoring real-world factors such as resistance, friction, heat loss, noise or material imperfections.

4, The thermal voltage V_T of diode is given by: $V_T = \frac{KT}{q}$

$K = 1.38 \times 10^{-23}\ J/K$

$q = 1.6 \times 10^{-19} C$

$T(K) = T(^\circ C) + 273 = 25 + 273 = 298K$

Apply the formula: $V_T = \frac{KT}{q}$

$V_T = \frac{(1.38\times10^{-23})(298)}{1.6\times10^{-19}} = \frac{1.38\times298\times10^{-23}}{1.6\times10^{-19}}$

$= \frac{411.24\times10^{-23}}{1.6\times10^{-19}} = 257.025\times10^{-4} = 0.0257V$

$= 0.026V = 26mV \quad \therefore V_T = \underline{26mV \text{ at } 25^\circ C}$

46 $I_D = I_S\left(e^{\frac{V_D}{nV_T}} - 1\right)$

$I_S = 40nA,\ n = 2,\ V_D = 0.5V,\ T = 25^\circ C \Rightarrow V_T = 0.026V$

$40nA = 40\times10^{-9}A$

$I_D = 40\times10^{-9}\left(e^{\frac{0.5}{2\times0.026}} - 1\right) \Rightarrow I_D = 40\times10^{-9}\left(e^{\frac{0.5}{0.052}} - 1\right)$

$I_D = 40\times10^{-9}\left(e^{9.615} - 1\right) \quad e^{9.615} \approx 15{,}000$

$I_D = 40\times10^{-9}\times15000 \Rightarrow 600000\times10^{-9} = 6\times10^{5}\times10^{-9}$

$\Rightarrow 6\times10^{-4} = 6\times0.0001 = 0.0006A \Rightarrow 0.6mA$

$I_D = \underline{0.6mA}$

What I Learned:

I, learned how to apply semiconductor theory to practical calculations, particularly determining thermal voltage $V_T = \frac{KT}{q}$ and computing diode current using diode equation. I gained a better understanding of the role of temperature in semiconductor behavior and how exponential relationships significantly impact diode current.

The main difficulty I encountered was managing scientific notation and evaluating exponential expressions accurately. Coverting temperature from Celsius to Kelvin and simplifying large numerical results required careful step-by-step calculation. However, working through each stage systematically improved my understanding of both the mathematical and physical concept involved.